KUSH PATEL

Data Analyst | BI & Analytics Engineer | Power BI · SQL · Python · Microsoft Fabric · Databricks · dbt

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Open to remote and relocation across Canada · Legally eligible to work in Canada · Available immediately

SUMMARY

- Data analyst and BI / analytics engineer with 4+ years turning raw operational and financial data (ERP, POS, WMS, workforce and REST feeds) into governed warehouse layers, Power BI reporting and decisions that Sales, Finance and Supply Chain can defend. Sole analyst across a perishable-goods supply chain; before that, a financial analyst running month-end GL/P&L reconciliations.
- Builds dimensional models on cloud warehouses (Microsoft Fabric, Azure Synapse, SQL Server) with Bronze/Silver/Gold medallion layers, conformed dimensions and SCD Type 1/2, orchestrated in Azure Data Factory and Fabric pipelines (dbt and Airflow in project work, not production).
- Delivers Power BI end to end — semantic model, DAX measure library, RLS, publish-ready reports — and treats data quality as a control: automated reconciliation cut recurring discrepancies an estimated ~45% by fixing issues at the feed rather than in the report.
- Twenty-one public projects carry **13,921 automated tests** in CI, fifteen with live demos, spanning BI, marketing measurement and revenue reconciliation, HubSpot CRM development, web analytics, analytics engineering, data engineering, forecasting, experimentation and optimization, anomaly detection, text-to-SQL and business-analysis delivery (requirements, UAT, business cases).

TECHNICAL SKILLS

BI & Data Visualization: Power BI (DAX, Power Query/M, semantic models, row-level security (RLS/OLS), incremental refresh, TMDL/PBIR), SSRS paginated reports, Tableau, Excel (Power Pivot, dynamic arrays, LAMBDA, what-if data tables), Streamlit, Plotly

Languages: SQL (T-SQL, window functions, CTEs), Python (pandas, NumPy, scikit-learn, Flask, FastAPI), PySpark, DAX, M

Warehousing & Lakehouse: Microsoft Fabric (Lakehouse, Warehouse, OneLake, Direct Lake), Azure Synapse Analytics, SQL Server, PostgreSQL, Databricks (SQL warehouse, serverless jobs, Unity Catalog), DuckDB, Delta Lake, Snowflake (dbt profile, not yet run)

ETL / ELT & Orchestration: Azure Data Factory, Fabric Data Pipelines & Dataflows Gen2, dbt Core, Apache Airflow, SSIS, SQL Server Agent, REST API integration, incremental and CDC-style loads, schema-drift contracts, quarantine-and-replay

Modelling & Governance: Kimball star schemas, conformed dimensions, SCD Type 1/2, metric catalogues with owner and grain, data lineage, RLS/RBAC and column masking, de-identification (k-anonymity), reconciliation and control totals, cutover validation

Analytics & ML: forecasting with rolling-origin backtests, classification (XGBoost, logistic regression), anomaly detection, survival analysis, recommender systems, randomized experiments and causal uplift, mixed-integer optimization (HiGHS, Gurobi), geo holdouts and difference-in-differences, LLM text-to-SQL; always graded against a simple baseline

Business Analysis: stakeholder engagement, requirements and acceptance criteria, current/future-state mapping, requirements traceability, UAT, business cases and options analysis, RACI/RAID, change and adoption planning

Quality & Delivery: pytest, dbt tests, pandera contracts, referential-integrity checks, anomaly thresholds, Git, GitHub Actions, Docker

PROFESSIONAL EXPERIENCE

Two Rivers Specialty Meats

Dec 2023 – Jun 2026 · North Vancouver, BC

Data Analyst — Operations, Logistics & Enterprise Reporting

- Sole analyst for a perishable-goods distributor: integrated ERP, Square POS, workforce and REST API feeds into curated Bronze/Silver/Gold datasets serving ad hoc analysis, BI dashboards and operational reporting.
- Designed and built the enterprise data mart applying star schemas, conformed dimensions and slowly changing dimensions, producing reliable data products sliced by SKU, supplier, region, customer and channel.
- Owned Power BI end to end — semantic model design, a standardised DAX measure library, drill-through and publish-ready reports — giving Sales, Finance and Supply Chain self-service access without analyst dependency.
- Engineered cloud-native ELT in Microsoft Fabric and Azure Data Factory/Synapse using SQL and PySpark, with incremental loads, parameterised transforms and structured error handling.
- Built reconciliation controls comparing source against curated totals for GL/P&L and inventory valuation, cutting discrepancies an estimated ~45% and reporting errors ~30% by resolving issues at the feed level rather than patching reports (internal figures, directional rather than audited).
- Maintained data quality across the mart: automated schema, completeness, uniqueness and referential-integrity tests, anomaly thresholds, exception logging and pipeline alerting.
- Implemented FEFO and lot-level traceability for recall readiness, delivered customer intelligence reporting (inactivity signals, lost-account tracking, rep coverage), and built SKU-level demand forecasting deployed as Python web apps that reduced stock-outs and perishable waste.
- Owned the legacy SQL Server estate through migration: maintained SSIS packages and SSRS paginated reports while refactoring priority pipelines to Fabric notebooks, validating cutover with row counts, checksums and control totals so no reporting was disrupted.
- Partnered with Sales, Finance and Supply Chain to define governed KPIs (revenue, COGS, gross margin, OTIF, inventory turns, days on hand) with documented definitions and acceptance criteria, secured with RLS/RBAC and column masking as self-serve access widened.

Shivam Investments Dec 2020 – Jul 2022 · India / Remote

Financial Analyst — Data & Reporting

- Gathered reporting requirements from portfolio and risk stakeholders and delivered Power BI dashboards with drill-through and auditable measure definitions.
- Automated recurring analysis with Python (pandas) and SQL views feeding Power BI, replacing manual spreadsheet consolidation and cutting manual effort an estimated ~40%.
- Performed GL/P&L reconciliations across sources and investigated discrepancies to maintain finance-grade accuracy.

SELECTED PROJECTS — public code, live demos and CI-tested results

Retail Analytics Platform — Python, Flask, DuckDB, data products & metric governance · 1,431 tests
github.com/KushPatel29/wholesale-analytics-platform · live: kushpatel29.github.io/wholesale-analytics-platform
BI platform and metric governance: Four dashboard pages agreed on revenue to the cent while margin did not; one cause was $333,573 of cost charged to 3,056 order lines that never shipped. One catalogue of 62 governed metric definitions (47 implemented, 15 withheld with the reason stated) now sits behind every page, nine accountable data products carry 27 quality gates, and the KPI diff that exposed the gap fails the build.

Supply Chain Control Tower — Microsoft Fabric patterns run on Databricks, PySpark, Delta Lake, Power BI, Streamlit · 836 tests
github.com/KushPatel29/supply-chain-control-tower · live: kush-network-risk-decision-room.streamlit.app
Data platform and operations analytics: A medallion pipeline defended in three layers — schema contracts before Bronze, row quarantine with replay, a quality gate on Gold — each proven in CI by injecting the failure it exists to catch, and benchmarked at 10M rows in Delta Lake. The eight-page Power BI control tower shows short-shipping, not lateness, is the bigger OTIF driver; a live studio routes disruption scenarios to owner-assigned actions.

Ask Your Data — Python, DuckDB, LLM text-to-SQL, Streamlit, voice · 1,163 tests
github.com/KushPatel29/ask-your-data · live: ask-your-data-kp.streamlit.app
Grounded text-to-SQL: Ask 71 tables across eleven business domains a question, typed or spoken, and every answer comes from a query that actually ran, with the SQL shown. A read-only guard and default-deny access policy sit in front of the warehouse; on a 58-question contract the keyless compiler scores 46 right, 0 wrong and 12 refused rather than guessed.

GL / P&L Reconciliation — SQL, Python, Power BI, Tableau · 710 tests
github.com/KushPatel29/gl-reconciliation-dashboard
Month-end close controls: Classifies ERP-versus-subledger discrepancies, traces every dollar to source and tests all 5,400 journal entries against controls. A certification layer evaluates 60 account-periods against a versioned tolerance and SLA: 8 auto-certified, 9 routed to a named reviewer and 43 blocked, with segregation of duties enforced and no approval fabricated. The same engine reconciles a FOCUS-format cloud bill unmodified.

Finance Decision Models in Excel and Power BI — Excel, Power Query, Power Pivot / DAX, Power BI (PBIP) · 1,190 tests
github.com/KushPatel29/excel-fpa-model
Two decision models: Portland General Electric plan-vs-actual from EIA, FERC and NOAA, published in Excel and a seven-page Power BI report, with a backtested load forecast; and the preserved Kestrel Bay distributor workbook for channel economics and working capital.

B.C. Local-Government Finance — Python, dbt / DuckDB, Excel, Power Query · 181 tests
github.com/KushPatel29/bc-local-government-finance
Public-sector finance on official filings: 90 provincial files held to written contracts, a dbt model with 31 data tests, and an Excel plan for any of 161 municipalities. Found the 68 municipalities investing below renewal need are $570M a year short, and caught the Province's own data defects.

Inventory Analytics — Operations Decision Studio — Python, Flask, Plotly, DuckDB, SQL, Docker · 349 tests
github.com/KushPatel29/inventory-analytics-app · live: kushpatel29.github.io/inventory-analytics-app
Forecasting and inventory policy: A six-workspace planning app over a 420-SKU, seven-node network. Seven forecasting methods compete in rolling-origin backtests; a policy lab prices seven service targets on an annual cost frontier and screens network transfers before recommending a purchase, ending in 1,170 prioritized actions. A BA case traces 13 requirements and 12 UAT scenarios through to a pilot plan.

Canada Wildfire Risk — Python, XGBoost, GIS, Power BI, GitHub Actions · 111 tests
github.com/KushPatel29/wildfire-prediction-app · live: wildfire-prediction-app1.streamlit.app
Machine learning on real public data: A first-prize 2024 hackathon project rebuilt as a daily seven-day ignition forecast from 164,707 recorded fires and 3,198 weather stations. Two calibrated XGBoost models, scored once on unseen 2020–2024 seasons, reach ROC-AUC 0.881 and put 57.5% of fires in each day's riskiest tenth of cells. GitHub Actions rebuilds the forecast twice a day.

Corridor Decision Intelligence — Python, DuckDB, causal ML, HiGHS/Gurobi, Databricks, Power BI, Streamlit · 244 tests
github.com/KushPatel29/decision-intelligence-platform · live: corridor-decision-intelligence.streamlit.app
Offer targeting, experiments and optimization: Chooses which of 17,059 simulated toll-road customers get which of nine offers. Uplift models trained on a ten-arm randomized trial feed one mixed-integer program over 112,563 decisions, proven optimal. Scored against the simulator's truth, the plan reaches 75% of the perfect-knowledge ceiling, versus 7% for propensity targeting; a randomized test of the policy measured +$1.40 per customer.

EDUCATION

Northeastern University — Master of Professional Studies in Analytics (Applied Machine Intelligence), GPA 3.76/4.0 · Vancouver, BC — Sep 2022 – Mar 2024

Gujarat Technological University — Bachelor of Engineering, Computer Science · India — Aug 2017 – Aug 2021

AWARDS & CERTIFICATIONS

First prize, 2024 wildfire-prediction hackathon (now a live forecast) · Microsoft Certified: Fabric Analytics Engineer Associate (DP-600) — in progress · Google Data Analytics Professional Certificate · UC Davis SQL for Data Science · IBM Python for Data Science, AI & Development · Databricks Lakehouse Fundamentals · Power BI Data Modelling with DAX